

EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 12- 400 - 2

**THREE PHASE (LOW- LOSSES) AUTOTRANSFORMER UP TO
24 KV WITH ON LOAD TAP CHANGER**

Date: 23-07-2024

EDMS 12-400-2
SPECIFICATION
FOR
THREE PHASE (LOW- LOSSES)
AUTOTRANSFORMER UP TO 24 kV
WITH ON LOAD TAP CHANGER

Issue: Jul-2024 / Rev- 2

- This revision contains option items that must be selected by EDC before bidding.
- EDMS 11-400 for Surge Arrester shall be attached
- EDMS 25-401 for Mineral Insulating Oil shall be attached.
- EDMS 25-400 for Silica-Gel shall be attached.

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1. SCOPE

This specification describes the minimum technical requirements of design, engineering, manufacturing, inspection, testing and installation of outdoor, three-phase, oil immersed, naturally cooled (ONAN), low losses Autotransformer with on load tap changer.

This Autotransformer should be used to regulate the medium voltage MV distribution networks (12 kV and 24 kV, 50 Hz systems).

The Autotransformer should be equipped with on-load tap-changer (OLTC) to adjust the tapping steps in two-modes:

1- Manually

- a- Local operation (by handle or selector switch) from the tap-changer control cubicle.
- b- Remote operation from remote tap-changer control cubicle (RTCC).

2- Automatically

By using automatic voltage regulating instrument (AVRI) (**Optional according to ...EDC requirements**).

2. APPLICABLE CODES AND STANDARDS

Unless otherwise specified in this specification, this autotransformer with on-load tap changer should comply with the latest edition of IEC standards given in Table (1).

Table (1)

Standard No.	Description
IEC 60076 Power Transformers	Part 1 General
	Part 2 Temperature rise for liquid-immersed transformers
	Part 3 Insulation levels, dielectric tests, and external clearances in air
	Part 4 Lightning impulse and switching impulse testing
	Part 5 Ability to withstand short circuit
	Part 7 Loading guide for mineral oil immersed power transformers
	Part 10 Determination of sound levels
IEC 60137	Insulated bushings for alternating voltages above 1 kV
IEC 60214	IEEE C57.131: Tap Changer.
	IEEE C57.15: Step voltage regulator.
IEC 60296	Mineral insulating oils for electrical equipment
ISO 9001	International standard for a quality management system (QMS)

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3. ENVIRONMENTAL CONDITIONS

The performance of the required autotransformer with on-load tap changer should be guaranteed under the environmental conditions given in Table (2).

Table (2)

Ambient temperature	-5°C to +45°C (50°C as option)
Maximum relative humidity	95 %
Altitude	≤ 1000 meters.

4. AUTO-TRANSFORMER REQUIREMENTS

4.1 Rated operating voltage value

- The rated operating voltage value at no-load should be as follows: (or customizing the product according to the –EDC / customer requirement).
 - For 12 kV system voltage: from 8.325 kV to 11 kV.
 - For 24 kV system voltage: from 17.555 kV to 22 kV.

4.2 Tapping

1. The reference tap shall be (**center tap**) as follows: (or as per --EDC requirement).
 - For 12 kV system voltage: 9.625 KV
 - For 24 kV system voltage: 19.775 KV
2. The manual position selector of the OLTC: local (L) / remote (R).
3. The on-load tap-changer shall be provided with an automatic voltage regulating instrument (if required by –EDC / customer requirement) and a remote tap-changer control cubicle (RTCC).
4. Tapping range should be as follows: (or as per –EDC/customer requirement)
 - **For 12 kV System**
 - Autotransformer with nine tapping steps, each step equals (325 volts).
 - Tabs will be: 8.325, 8.650, 8.975, 9.300, **9.625**, 9.950, 10.275, 10.600 & 10.925 KV
 - **For 24 KV System:**
 - Autotransformer with nine tapping steps, each step equals (555 volts).
 - Taps will be :17.555, 18.11, 18.665, 19.22, **19.775**, 20.33, 20.885, 21.44, 21.995 KV

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4.3 Nominal current

- The nominal current of Autotransformer with on load tap changer should be at minimum input voltage: 100, 200, 300, 400, 500 A (or as perEDC requirement).
- The auto transformer should deliver the rated power continuously without exceeding the permissible temperature rise limits according to the latest IEC 60076-2.

4.4 Connection symbol

- The connection symbol of the auto transformer should be: (Y_{auto}. YN₀).

4.5 Impedance voltage

- The impedance voltage of the Autotransformer with on load tap changer at rated power should be less than 1.5 % of the rated voltage (11 kV or 22 kV).

4.6 No Load current

- The no-load current of the Autotransformer with on load tap changer should be not more than 0.5% of rated current.

4.7 Short-circuit duration

- The Autotransformer with on load tap changer should be designed and constructed to withstand the mechanical and thermal stresses produced by external short circuits without damage of at least 25 times the base (r.m.s) symmetrical rated load current
- The short circuit withstand duration should be not less than 2 seconds.

4.8 Overload capacity

- -The Autotransformer with on load tap changer should withstand permissible overload capacity of 20% for 1.5 hr. (without exceeding the permissible heating).

4.9 Efficiency and Losses

- This autotransformer with on-load tap changer should be designed for maximum efficiency and minimum losses.
- When comparing different tenders, the present worth (P.W.) value of the capitalized cost of guaranteed losses in the autotransformer with on-load tap changer should be added according to the following formula

$$P. W = K \times 8760 \times C [W_i + (L.S.F) P^2 W_{load}] L.E$$

Where

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$$K = [(1 + i)^{N-1} / i (i+1)^N]$$

K = Present worth factor (L.E) based on latest interest rate from central bank of Egypt over 20 year.

i: Latest interest rate from central bank of Egypt.

N: Number of years.

C = Cost of KWH.

(According to the max commercial cost of KWH and can be adjusted as per EEHC update)

LS.F = Loss factor for the load = 0.4.

W_i = No load (Iron) losses in KW at rated voltage and main tapping.

W_{cu} = Load (Copper) losses in KW at full load and 75°C .

P = Peak load (P.U.) = 0.8.

The values of the required losses are attached in table 1.

4.10 PENALTY AND REJECTION

4.10.1 Rejection of material

- If during inspection tests attended by the ...EDC representative engineer, any material is found to be defective or not manufactured according to the specifications agreed upon, the ...EDC representative engineer has the right to reject such material.
- In such case, the contractor should be obliged to replace the defective material without being entitled to any extra payment whatsoever or to any extension of time of completion of contract.

4.10.2 Penalty and Rejection for Decreased Efficiency

- The iron and the copper losses should be kept within the guaranteed values.
- The measured losses values should not exceed the allowable tolerances in IEC or approved equal standards.
- The ...EDC representative should have the right to reject the transformer with such losses which exceed the limits.
- Max. Permitted power losses of transformer (at 95°C) as the following table:

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Table (3)

Rated current (A)	12 kV System At reference point		24 kV System At reference point		Sound pressure level dB
	No-load losses (Watt)	Load Losses (Watt)	No -load losses (Watt)	Load Losses (Watt)	
100	576	3815	576	3815	48-56
200	825	6430	825	6430	48-56
300	1222	9450	1222	9450	48-56
400	1500	11700	1500	11700	48-56
500	1785	13860	1785	13860	48-56

4.11 Temperature –Rise Limits

- Autotransformer should be able to deliver its max. continuous ratings, without exceeding the permissible temperature rise limits “over the ambient temperature”:
 - Top oil temperature rise : 45°C.
 - Winding temperature rise measured by resistance method: 55°C.
 - Hot spot temperature: 60°C

4.12 Noise Level

- The core joints should be properly interleaved; precaution should be taken to keep the noise at the level of (48 – 56) dB according to rated capacity and measured at distance of (one) meter, at mid tap.

4.13 Harmonics level

The Autotransformer should be designed such that total harmonic distortion in voltage waveform THDV does not exceed 5%.

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5. CONSTRUCTION:

5.1 Core

- The core should be made of high permeability, low loss, silicon steel laminations.
- The laminations should be cold-rolled, grain-oriented with thickness not more than 0.3 mm.
- Each lamination should be insulated on both sides by any suitable method except by using solid insulation.
- The insulation of core lamination must endure heat and scratches and not affected by Transformer oil.
- The core should be constructed and assembled in such a way in order to have the minimum no-load loss, no-load current, noise, voltage harmonics, especially the third harmonics and to withstand the mechanical and thermal stresses.

5.2 Windings and Terminals

- The windings should be manufactured from high conductivity electrolytic copper, complying with the IEC standard in that the maximum flux density in the yoke should not be more than 1.6 tesla.
- The insulation of winding should be designed to withstand power frequency and impulse over voltage stresses safely.
- The line terminals should be brought out through suitable porcelain bushings.
- The bushings leakage path should be not less than 4.5 cm/kV of rated line voltage.

5.3 Surge Arresters

- The automatic regulated Autotransformer should be provided with 6 surge arresters to protect transformers against transient over voltage.
- Surge arresters should be zinc oxide and specified according to the latest applicable IEC standards.
- The surge arresters leakage path should be not less than 4.5 cm/kV of rated line voltage.

5.4 Protection

- The automatic regulated Autotransformer should be provided with protection system to get the Autotransformer out of service in case of failure of one or two phase by installing ring main (1LBS+1C.B) with measuring cell (scope ofEDC)

5.5 Tank

- The automatic regulated Autotransformer tank should be made of welded sheet steel or boiler plate steel with suitable thickness to withstand the different mechanical stresses occurring during operation or drying process of transformer insulation.

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- The tank cover should be bolted and sealed to the tank and provided with thermometer pocket, lifting lug, terminal bushing, connection pipe to the conservator and if necessary with pressure relief device, oil filling valve and air-taking plugs.
- The tank base should be fitted with oil drain and sample valve, earthing screws and pulling plates.
- The tank should be provided with suitable cooling radiator and automatic regulated Autotransformer nameplate.
- The Autotransformer should be designed such that it can be partially or completely untanked for inspection and maintenance without disconnecting any internal electrical or mechanical connections.
- The external parts of the tank and control enclosure should be painted gray according to ANSI -70. The inside of the tank and bottom of the cover should also be primed and painted.

5.6 Conservator

- Automatic regulated Autotransformer should have an oil expansion conservator fabricated from steel sheet of suitable thickness and should be connected to the tank through the connection pipe on the tank Cover.
- The oil conservator should be fitted with dehydrating orange silica-gel breather, oil level indicator, oil drainage plug and oil filling plug.

5.7 Accessories

- Thermometer.
- Buchholz relay
- Four wheels: if only requested byEDC and should be suitable to move the transformer in two perpendicular directions.
- Bypass disconnector switch (optional).
- One digital multi meter on each side (incoming & outgoing) measures all main variables (KV, I, KW,KVAR,etc.)

5.8 Transformer oil

- The autotransformer should be filled to the required level with new, unused, clean, standard mineral insulating oil compliance with IEC 60296.
- The oil should meet the requirements of the latest revision of the technical specifications EDMS 25-401.

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6. TOLERANCES

- The applied tolerances should not exceed the following values:
 - Rated voltage ratio $\pm 0.5\%$ of the specified values.
 - No-load current $\pm 30\%$ of the specified values.

7. ON-LOAD TAP-CHANGER (OLTC) REQUIREMENTS

1. The OLTC should be "on tank type " **or** "in tank type ".(as per EDC requirements)
2. Direction of operation indicator for Raise /Lower tap should be available.
3. Dynamic short circuit certificate should be provided with tender.
4. The technical specifications of the OLTC should be as follows:

Parameter	Description
No. of Phases	3
Type	"In-tank", or "on tank " Oil type
Tapping	Linear
No. of operating positions (taps)	9 (or as per customer requirement)
Voltage per step	325 V for 11kv system (or as per customer requirement) 555 V for 22kv system (or as per customer requirement)
Rated through current	According to the transformer rate
Time per tap-change	2-4 sec.
Rated short circuit withstand current	Up to 5 KA for 3 sec.

5. The OLTC should be provided with a motor driving unit integrated into the tap-changer housing and operated from electrical control cubicle.
6. The motor drive unit should contain a thermostatically controlled anti-condensation heater (≥ 50 W), hygostat, door-operated light and a manual handle for emergency and maintenance operation.
7. The Motor drive unit should be weatherproof ($\geq$ IP54).

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8. Tap Changer operating mechanism motor should be 3-phase, 400V, and 50 HZ supplied through the auxiliary supply (sub coils) from the autotransformer itself or any other suitable supply.
9. The relays, switches and push buttons should be enclosed in a weatherproof cubicle ($\geq$ IP54) and should be readily accessible for maintenance at ground level with the transformer energized.
10. The on-load tap changer should be operated as follows:
 - a. Locally from the tap-changer control cubicle or from the remote terminal control cubicle (RTCC).
 - b. Automatically using automatic voltage regulation (AVRI) (optional).
11. The OLTC should be provided with the following:
 - a. A.V.R with digital screen.
 - b. Pressure relief valve.
 - c. Remote control cubicle RTCC.
 - d. Voltage indicator.
 - e. Tap position indicator.
 - f. Potential transformer for supplying the A.V.R

12. Automatic Voltage Regulating (AVR) (optional)

- a. The RTCC should be customized as per the needs of the --EDC requirements for various indications that are required. Further customization of annunciation and SCADA compatible RTCC should be also possible.
- b. This AVR should consist of the following:
 1. Raise and lower voltage set point (90-120V)
 2. Time delay for command (30-120 sec) (adjustable)
 3. Dead band to avoid spurious operations (2 to 9%)
- c. The AVR should execute the tap change command if the voltage crosses the set point and remains there till the time delay has elapsed.
- d. Additional features of under voltage/ over voltage blocking, line drop compensations should be provided.
- e. The AVR should be able to be protected with the necessary protection system to avoid over & under voltage, reverse power, and over current. (--EDC scope).

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8. MARKING

- Each automatic regulated autotransformer should be provided with an easily readable nameplate of weatherproof material fitted in a visible position.
- On the nameplate, the following data should be indelibly marked (by etching engraving or stamping)
 - a. Manufacturer's name.
 - b. Manufacturer's serial number.
 - c. Year of Manufacture.
 - d. Number of phases.
 - e. Rated frequency.
 - f. Rated power.
 - g. Rated voltage.
 - h. Rated current.
 - i. Connection symbol.
 - j. Impedance voltage at rated current.
 - k. The voltage values at all tapping positions.
 - l. The specification applied to the transformer.
 - m. S.C. withstands current and duration.
 - n. Type of cooling.

9. INSPECTION AND TESTING

9.1 General

- The tenderer should submit the routine and type test certificates.
- --EDC reserves its right to carry out inspection during fabrication stages and witness the testing of the transformers.
- Unless otherwise specified or approved in writing by --EDC, all transformers should be tested at the tenderer's factory and on his expense in accordance with the latest applicable relevant IEC standards, under control and supervision of --EDC inspection engineer.
- Attendance of --EDC representative during fabrication and testing stages should not relieve the tenderer of his full responsibility for furnishing the transformers in compliance with this specification requirements and should not give him the right to invalidate any claim which --EDC may issue because of defective or unsatisfactory

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material or faulty workmanship.

- The tenderer should submit a quality control procedure with his offer.
- The tenderer should be obliged to replace the defective material without being entitled to any extra payment or to any extension in the contract period.

9.2 Autotransformer Routine Tests

- The routine tests should be applied to each transformer in accordance with the requirements of IEC standards for transformers, these tests should include but not be limited to the following:
 - a. Measurement of D.C winding resistance.
 - b. Turns Ratio and polarity check or vector-group symbol.
 - c. Measurement of impedance voltage.
 - d. Measurement of load loss.
 - e. Measurement of No-load loss.
 - f. Induced over-voltage withstand test.
 - g. Separate-source power frequency voltage withstand test.
 - h. Electrical oil tests.
 - i. Insulation tests between winding and earth, and between windings.

9.3 Autotransformer Type Tests

- These tests should be carried out on one automatic regulated autotransformer for each type in accordance with the requirements of IEC standards for transformers, these tests should include but not be limited to the following:
 - a. Full wave lightning impulse voltage withstand test.
 - b. Temperature rise test.
 - c. All tests according to IEC 60296.
 - d. Short circuit test.
 - The supplier shall submit a short circuit test in an approved lap by the EEHC otherwise the supplier shall comply with the following:
 - Short circuit calculations for the autotransformer need approval shall be submitted to the Extra High Voltage Research Center for review with the same sample autotransformer with its drawing.

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- The supplier shall guarantee the autotransformer for (3) years, the guarantee remains valid throughout this period or till the short circuit test is submitted whichever comes first.
- The supplier will get approval for only one year and it will be renewed annually if there is no complaint from any EDC during the guarantee period (3 years).

10. SUBMITTALS

10.1 Submittals required with the tender:

- The tenderer should fill up and return a copy of the Technical Data Schedule.
- The tenderer should provide the relevant catalogues of the offered equipment/material and its required accessories, these catalogues should include layout, dimensions, technical parameters, drawings, and instructions for erection & operation & maintenance requirements.
- The latest valid approval certificate issued by Egyptian Electricity Holding Company.

10.2 SPARE PARTS (OPTIONAL)

- A comprehensive list of the manufacturer's recommended spare parts should be included in the tender.
- The quantities offered should be adequate for the initial five (5) years of operation.
- Spares supplied should be packed to provide long storage without deterioration.
- Each package should be clearly marked in Arabic and English with a description of its contents, if any spare part requires special storage conditions, these conditions should be detailed.
- --EDC has the right to request any spare part within five years of the operation date, and the tenderer should comply with this condition.

11. GUARANTEE

- The tenderer should guarantee the autotransformer against all defects arising out of faulty design, workmanship, or of defective material for a period of 12 months from putting in service or 18 months from the delivery date.

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12. TECHNICAL DATA SCHEDULE

- The tenderer should fill in the attached technical data schedule. Any offer not accompanied by a clear and complete technical data schedule shall be rejected

TECHNICAL DATA SCHEDULE AUTOTRANSFORMER WITH ON-LOAD TAP CHANGER

Particulars	
1- Manufacturer's name	
2- Type	
3- Applicable Standards	
4- Rated output (kVA)	
5- Maximum output (kVA)	
6- Method of cooling	
7- Connection symbol (group):	
- HV winding	
- LV winding	
8- Insulation	
9- Tapping switch (on HV side)	
- Number of steps	
- Value of the step	
- Range of regulation	
10- Material of core	
11- Thickness of core lamination (mm)	
12- Material of tank	
13- Thickness of tank	
- Sides (mm)	
- Cover (mm)	
- Bottom (mm)	
14- Weight of core (kg)	
15- Weight of winding (kg)	
Performance	

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16- Continuous output rated current at reference point (kVA)	
17- Permissible overload and duration	
- 10 % overload (minutes)	
- 20 % overload (minutes)	
- 30 % overload (minutes)	
- 50 % overload (minutes)	
18- No- load ratio	
- H.V (V)	
- L.V (V)	
19- No- load current	
- H.V side (A)	
- L.V side (A)	
20- Rated full load current with rated ratio	
- H.V side (A)	
- L.V side (A)	
21- Resistance per phase at 95°C	
- H.V side (Ω)	
- L.V side (Ω)	
22- Impedance voltage (%)	
23- Regulation at full load	
- Unity P.F (%)	
- 0.8 P.F (%)	
24- Max. Core flux density at full voltage and 50c/s (Tesla)	
25- Short-circuit withstand duration (sec)	
26- Max. Temperature rise above 45°C ambient temperature	
➤ At rated output	
- At top level (oil) °C	
- Winding °C	
- Core °C	
➤ At 20 % overload for 1.5 hours	
- At top level (oil) °C	

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- Winding °C	
- Core °C	
27- load losses at full load current and 95°C winding temperature (watts)	
28- No-load losses (watts)	
29- Total losses (watts)	
30- Efficiency at rated voltage ratio and 95°C	
➤ At 100% rated output and P.F = 1	
➤ At 100% rated output and P.F = 0.8	
➤ At 75 % rated output and P.F =1	
➤ At 75 % rated output and P.F = 0.8	
➤ At 50 % rated output and P.F = 1	
➤ At 50 % rated output and P.F = 0.8	
➤ At 25 % rated output and P.F =1	
➤ At 25 % rated output and P.F = 0.8	
31- Maximum permissible noise level at distance (1m/3m) (db.)	
32- One-minute power frequency withstand voltage, r.m.s (H.V winding/ L.V winding) (kV)	
33- Induced voltage withstand test (kV)	
34- 1.2/50 μ s Impulse withstand voltage, peak (kV)	
OLTC technical data	
35- No. of Phases	
36- Type	
37- Applied standards	
38- No. of operating positions	
39- Voltage per step (V)	
40- Rated through current (A)	
41- Time per tap-change (sec)	
42- Rated short circuit withstand current (kA)	
43- Rated duration of short-circuits (sec)	
44- Housing	
45- Material	
46- Dimensions (Length x Width x Height) (cm x cm x cm)	

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47- IP	
48- Weight (kg)	
49- Driving motor	
50- Power (watts)	
51- Voltage (V)	
52- Frequency (Hz)	
53- Ambient Temperature °C	
54- AVR Setting ranges	
- Desired Voltage Level (V)	
- Bandwidth (%)	
- Delay time T1 (sec)	
- Delay time T2 (sec)	
- Under voltage blocking (%)	
- Over voltage detection (%)	
- Over current blocking (%)	

I/we guarantee the correctness of above data for the autotransformers we are offering to Electricity Distribution Company

Signature of tenderer:

Date: / /